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PERSPECTIVE

Earth's dimming mirror: cloud contraction and the rise of albedo governance

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Abstract

Earth has reflected progressively less solar radiation since the start of the satellite era (1979), amplifying the planetary energy imbalance. Satellite analyses attribute the trend primarily to circulation-driven contraction of reflective cloud regimes, whereas process-based studies emphasize surface reflectivity changes. We argue that these perspectives are dynamically linked: as cloud cover contracts, diminishing atmospheric masking elevates the realized climatic leverage of surface albedo. This coupling makes surface reflectivity an increasingly important and manageable regulator of solar absorption under clearer-sky conditions. We propose an integrated governance framework that explicitly incorporates albedo management alongside carbon management.

1. A growing energy imbalance and the need for a coherent surface–albedo framework

Satellite observations from the Clouds and the Earth's Radiant Energy System (CERES) indicate that the Earth's energy imbalance—the net difference between absorbed solar energy and outgoing flux—has approximately doubled since the early 2000s, increasing from ∼0.5 to ∼1.0 W m^{-2} [1]. Independent ocean-heat content records confirm this acceleration, documenting an exceptional rise in planetary heat uptake since the onset of the satellite era in 1979 and the advent of high-precision CERES monitoring in 2000 [2, 3]. Early multi-decadal heat inventories established a baseline planetary imbalance [4] that has been decisively exceeded during the 1993–2023 period. Contemporary observations suggest that this amplification is partly associated with the progressive unmasking of greenhouse forcing as anthropogenic aerosol cooling declines [5]. A central implication of this 'clearing-sky' phase (characterized by declining aerosol optical depth, reductions in low-cloud cover, and cryosphere reflectivity loss) is that a larger fraction of incoming shortwave radiation now reaches the surface. As a result, surface albedo exerts increasing climatic leverage as reflective cloud cover contracts. Although the global imbalance remains the primary metric for tracking the climate trajectory, its spatial expression is highly heterogeneous and can be amplified at the biome scale. In grassland systems, for example, the imbalance has intensified from ∼0.8 to over 1.7 W m^{-2} (figure 1). Together, these observations document a transition from the near-equilibrium state of the 1970s toward a regime of sustained and amplified energy accumulation. These regional anomalies coincide with a pronounced 'clearing-sky' phase associated with declining aerosol optical depth, reductions in low-cloud cover, and shifts in land-surface properties that together enhance surface shortwave absorption [1, 5, 6]. Additional contributions arise from cryosphere reflectivity with sea-ice and snow loss [7].

Despite broad agreement on the observed decline in planetary reflectivity, attribution remains fragmented. One line of research emphasizes large-scale atmospheric reorganization and low-cloud feedbacks as the dominant driver of increased solar absorption, whereas another highlights the climatic

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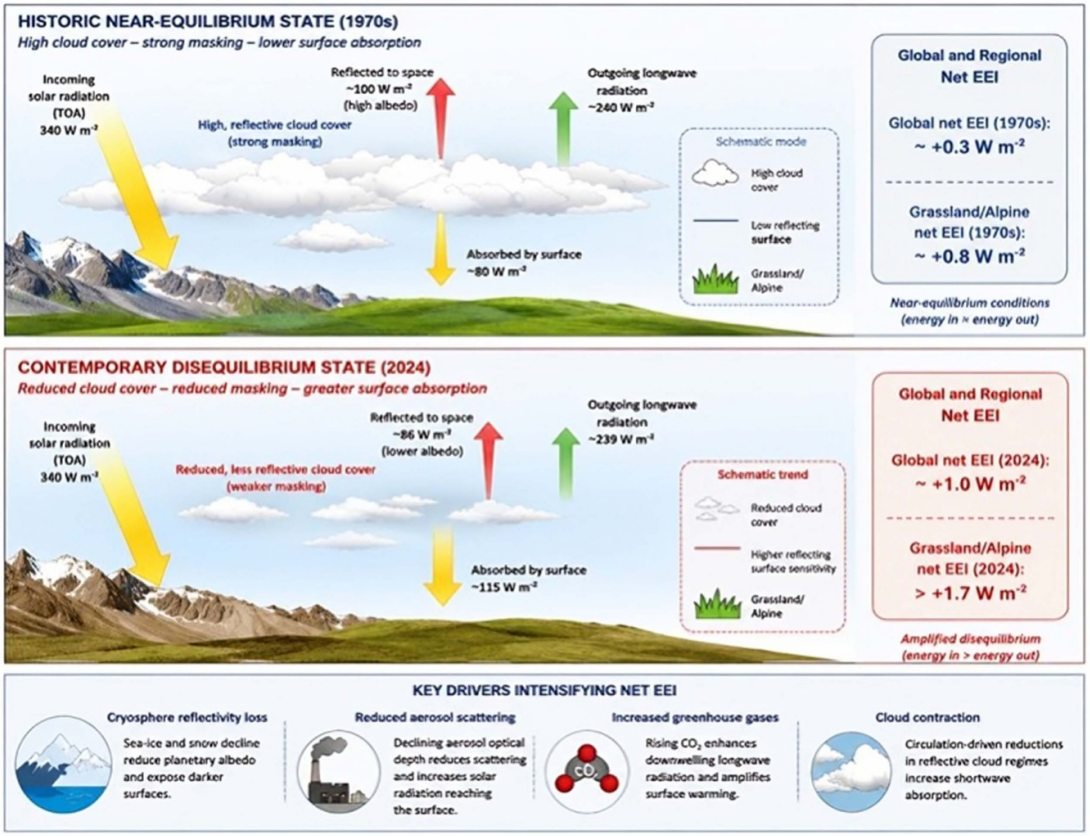


Figure 1. Transition from near-equilibrium to amplified Earth's energy imbalance (EEI) (1970s-2024). Schematic contrast between the historic near-equilibrium state (top) and the contemporary disequilibrium state (bottom), illustrated for grassland/alpine systems. The incoming solar radiation (yellow arrows) is held constant at the top of the atmosphere (TOA) across both panels to highlight that the increase in net energy imbalance is driven by internal atmospheric and surface modifications, rather than changes in external solar forcing. The bottom panel illustrates the intensification of net EEI through interacting mechanisms, including contraction of reflective cloud systems, cryosphere reflectivity loss, reduced aerosol scattering, and increased greenhouse-gas concentrations, which together enhance net shortwave energy absorption. Global net EEI increased from approximately $+0.3$ W m^{-2} in the 1970s to $\sim+1.0$ W m^{-2} by 2024, with regional amplification reaching over $+1.7$ W m^{-2} in grassland systems. Data synthesized and adapted from Loeb *et al* [1–3], von Schuckmann *et al* [2], and Hansen *et al* [4, 5]. Concept based on a user-provided sketch and scientific specifications supplied by the authors, with figure composition, layout, and graphical refinement assisted by generative AI tools (Gemini/Google AI and ChatGPT/OpenAI).

importance of land-surface reflectivity and cloud-surface interactions [6]. Recent satellite observations from the Amazon further demonstrate that cloud-mediated increases in top-of-atmosphere (TOA) albedo associated with forest loss can approximately double the cooling effect relative to surface brightening alone, underscoring the importance of coupled cloud–surface feedbacks in Earth's energy balance [8]. Here we argue that these perspectives are dynamically linked: as cloud cover contracts, a larger fraction of incoming shortwave radiation reaches the surface, increasing the climatic leverage of surface albedo and land-surface processes under clearer-sky conditions.

2. Cloud-regime contraction: a leading yet incomplete explanation

Satellite-based analyses provide compelling evidence for a top–down dynamical driver. Consistent with this view, Tselioudis *et al* [9] decomposed the global shortwave cloud radiative effect trend ($+0.45 \pm 0.09$ W m^{-2} decade^{-1} over the period 2001–2024) and showed that the resulting contraction of high-reflectivity cloud regimes accounts for $\sim$0.37 W m^{-2} decade^{-1}, or roughly 80%, of the observed increase in absorbed solar radiation. Crucially, Tselioudis *et al* [9] identified this contraction not as an isolated mechanism, but as the outcome of specific large-scale dynamical shifts: the narrowing of the intertropical convergence zone and the poleward shift of the mid-latitude storm tracks.

Rather than challenging the well-established distinction between high-cloud longwave warming and low-cloud shortwave cooling, this perspective highlights how recent dynamical shifts have systematically reduced the prevalence of the most reflective low-cloud regimes, thereby enhancing shortwave absorption under clearer-sky conditions. As Tselioudis *et al* [9]. conclude, 'the storm cloud contraction is found to be the main contributor to the observed increase of the Earth's solar absorption during the 21st century' (p. 1). This contraction is most pronounced in the mid-latitude storm tracks of both hemispheres

and the expanding subtropics, where the poleward migration of cloud systems exposes larger areas of the underlying ocean and land surfaces to incoming solar radiation. In doing so, cloud contraction does not replace surface controls but increases their radiative influence by reducing atmospheric masking and enhancing the climatic leverage of surface albedo [1, 6, 9].

3. Surface albedo: the overlooked yet strategic climate lever

If the atmosphere acts as the radiative 'shutter' of the Earth system, the surface functions as its primary 'absorber'. Gottschalk *et al* [10] show that land-use and management practices routinely alter surface albedo by 0.05–0.20, magnitudes capable of producing substantial regional radiative forcing. Surface reflectivity spans a diverse spectrum, from dark wet soils (∼0.05) and forests (∼0.05–0.15) to grasslands (∼0.16–0.30), croplands (∼0.15–0.25), and dry sands (∼0.45). High-latitude and high-altitude regions are dominated for much of the year by fresh snow (∼0.90), while urban environments range from ∼0.10 to 0.28, with high-albedo cool-roof coatings reaching ∼0.5–0.6. In contrast to stochastic atmospheric circulation, surface reflectivity is directly manageable and, therefore, policy-relevant. Its climatic importance thus depends not only on its intrinsic reflectivity but also on the degree of atmospheric masking by clouds.

Practices such as cover cropping, residue retention (the practice of leaving crop stalks and leaves on the field after harvest to act as a light-colored protective layer), and the deployment of high-albedo urban materials can measurably increase surface reflectivity. However, these interventions are not unambiguously beneficial, and a central paradox remains as stated by Gottschalk [10]: 'biogeophysical effects on climate associated with changes in surface albedo do not necessarily have the same sign, i.e. cooling or warming' (p. 2). These effects often oppose one another. For example, afforestation may sequester carbon (a biogeochemical cooling) while simultaneously darkening the surface and enhancing solar absorption (a biogeophysical warming). This tension highlights the need for a radiative-equivalent framework in which carbon management and albedo governance are evaluated jointly rather than in isolation.

4. Echoes from deep time: Mediterranean proxies for climate sensitivity

Multi-centennial reconstructions of Mediterranean cloudiness provide essential contexts for interpreting recent trends. Diodato *et al* [11] reconstructed western Mediterranean cloudiness over the 969–2022 CE period, revealing a gradual increase in cloud cover throughout the early part of the second millennium. This trend culminated in a cloud cover peak around 1600 CE (coinciding with the coldest phase of the Little Ice Age), before entering a persistent long-term decline toward unprecedented modern minima. A complementary reconstruction (1500–2022 CE) across the broader Mediterranean [12] shows a progressive decline beginning in the 17th century, followed by a post-early-19th century shift toward greater multidecadal variability and sustained lower levels. Notably, this transition began during the later phase of the Little Ice Age, indicating that the onset of cloud decline preceded the significant anthropogenic greenhouse forcing. This early onset likely reflects a synergistic unmasking of the land surface, driven by the combined effects of natural variability and historical land-use changes. Natural contributions include post-volcanic recovery following major eruptions (e.g. Tambora in 1815), periods of elevated solar activity, and positive phases of the Atlantic multidecadal oscillation (AMO). On the land-use side, peak European deforestation between approximately 1750 and 1850 reduced forest cover, which altered regional albedo, decreased evapotranspiration, and weakened moisture recycling. The resulting decline in convective cloud formation thinned the atmospheric cloud shield, exposing the surface to greater incoming solar radiation well before greenhouse-gas forcing intensified [13, 14].

Satellite-era observations (1982–2022) refine this historical picture. They reveal a marked decline in annual cloud cover alongside increasing temperatures, though with significant regional heterogeneity. Hotspots such as Sicily and northern Tunisia exhibit reductions approaching 10% per decade, coincident with a systemic shift around 2001. This period is characterized by altered spatial structures, variance distributions, and evolving connections to North Atlantic circulation anomalies (thermohaline processes). Conversely, previously reported heliospheric associations (e.g. links to galactic cosmic rays) have become heterogeneous [15]. These changes suggest a reorganization of coupled ocean–atmosphere variability, potentially amplified by anthropogenic forcing but not solely driven by it.

These long-term declines unfold against a backdrop of well-established natural oscillatory influences. Solar variability and large-scale modes such as the AMO contribute to the ∼60 year multidecadal signals evident in Mediterranean cloudiness. Planetary gravitational harmonics may also modulate solar activity in ways that align with these quasi-periodic patterns, though the underlying mechanisms remain debated [16]. Yet it is against this oscillatory background that a striking paradox emerges: the Mediterranean climate system is now behaving in ways inconsistent with historical controls. As Diodato *et al* [11] noted, 'recent decades have witnessed an

unprecedented coupling of intense solar activity, high temperatures, and the lowest cloud cover on record' (p. 1). Since ∼2001, cloud dynamics appear increasingly decoupled from historical oscillations and more strongly governed by large-scale circulation shifts [9, 15]. This points to a transition from quasi-periodic variability toward a sustained structural reduction in reflective cloud cover, with far-reaching implications for shortwave albedo feedbacks, regional climate sensitivity, and future warming trajectories.

5. Coupled controls on radiation: from top–down to bottom-up

The apparent separation between circulation-driven change and internal variability dissolves when the system is viewed as a set of coupled controls operating across three interacting layers (figure 2), where circulation-mediated cloud changes arise from interacting layers, radiative adjustments, and modes of variability [17]. First, large-scale atmospheric circulation (top–down) shapes the spatial extent, persistence, and regime transitions of cloud fields [9]. Second, aerosol-cloud microphysics (intermediate) modulate cloud optical thickness and radiative efficiency [11]. Third, surface albedo (bottom-up) determines the fraction of incoming solar flux that is ultimately absorbed or reflected [10].

Together, these layers form a tightly coupled radiative system in which circulation sets the stage, microphysics tunes the optical properties, and surface reflectivity modulates the final partitioning of absorbed energy. This integrated perspective is essential for diagnosing recent anomalies and anticipating evolving climate sensitivity. In the intermediate layer, and considering only cloudy-sky conditions, we distinguish two regimes. Optically thick clouds (typical of cloudy skies with high aerosol loading or dense cloud columns) enhance shortwave reflection and produce a net cooling effect. Optically thin clouds (occurring under reduced aerosol loading or in thinner cloud layers) permit greater transmission of solar radiation to the surface. Clear-sky conditions follow different physical principles: there, aerosol optical depth and molecular scattering govern transmission independently of cloud processes [18], and are thus treated separately from the cloudy-sky optical thickness comparison above. For example, marine stratocumulus clouds exert a strong net cooling effect by efficiently reflecting incoming solar radiation, while deep convective clouds tend to produce a net warming effect by trapping outgoing longwave radiation more effectively than they reflect shortwave radiation [17]. As cloud cover declines, a larger fraction of incoming solar radiation reaches the surface, increasing the radiative leverage of surface reflectivity. Mediterranean observations illustrate how this shift amplifies nonlinear feedbacks: reduced cloudiness intensifies surface heating, alters evapotranspiration patterns, and reshapes boundary-layer dynamics. Specifically, this can trigger a proposed 'humidity paradox', consistent with established land–atmosphere coupling mechanisms. In this regime, CO_2-driven land-atmosphere interactions promote boundary-layer drying that inhibits cloud formation over land, reinforcing a positive feedback between reduced cloudiness, increased solar absorption, and perturbations to the surface water balance [19]. Although cloud attenuation varies across wavelengths, clearer-sky conditions generally increase the transmission of incoming shortwave radiation, enhancing surface absorption [20].

This asymmetry strengthens net surface energy gain without requiring strong spectral selectivity in cloud transmission. In this regime, wavelength-dependent surface emissivity may play a secondary but increasingly relevant role in regulating outgoing longwave radiation, particularly under dry and cloud-free conditions [21]. However, under increasingly clear-sky conditions, surface albedo becomes an increasingly influential control on the shortwave component of the planetary energy balance because diminishing cloud masking enhances the radiative sensitivity of the underlying surface. At the planetary scale, however, changes in cloud properties remain a major driver of TOA shortwave variability, particularly over oceans where surface albedo varies relatively little. Thus, cloud contraction and surface albedo should be viewed as complementary, dynamically coupled controls rather than competing explanations. As cloud cover declines, the climate system moves toward a state in which relatively small changes in surface albedo exert a disproportionately large influence on the surface energy balance. Order-of-magnitude estimates suggest that a 1% absolute increase in global planetary albedo could correspond to a cooling on the order of ∼1 °C [22], although the exact sensitivity depends on atmospheric state, cloud feedbacks, and spatial heterogeneity. Consistent with this framework, recent attribution studies further show that regional changes in cloud properties are often the primary driver of TOA shortwave albedo variability, while surface reflectivity modulates the realized radiative response under clearer-sky conditions [23].

6. Synergies, trade-offs, and policy relevance

Multiple land-atmosphere processes can combine additively, but their joint effects are rarely linear. Synergies are common: cover crops can simultaneously enhance surface reflectivity and sequester carbon [24], and cool roofs mitigate both urban heat islands and building energy demand [25]. Yet trade-offs remain highly context-dependent. Afforestation on naturally bright soils may lower albedo and

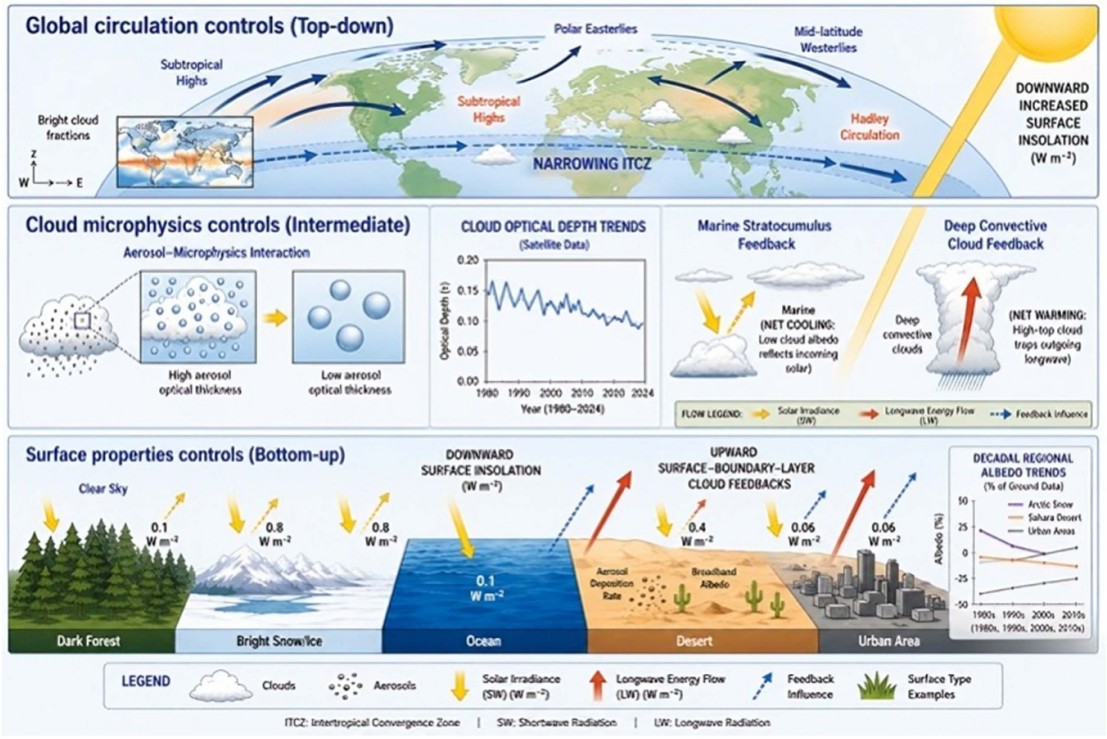


Figure 2. Coupled controls on Earth's shortwave radiation budget. The climate system operates as a dynamically coupled continuum across three interacting layers: (top) large-scale atmospheric circulation governs the spatial distribution of cloud fields; (middle) aerosol-cloud microphysics modulates cloud optical thickness (here referring exclusively to cloudy-sky conditions) and radiative efficiency; and (bottom) surface characteristics regulate the final partitioning of absorbed solar radiation [10]. Clear-sky radiative transfer is addressed separately in the text. As cloud cover declines, the relative radiative influence of the underlying surface intensifies through reduced atmospheric masking. The figure synthesizes key processes discussed in Tselioudis *et al* [9], Gottschalk *et al* [10], together with related satellite observations of cloud optical depth and regional albedo trends. Concept based on a user-provided sketch and scientific specifications supplied by the authors, with figure composition, layout, and graphical refinement assisted by generative AI tools (Gemini/Google AI and ChatGPT/OpenAI).

increase net absorption, while organic amendments or slurry applications can transiently reduce grassland reflectivity by more than 60% [10]. These interactions highlight the need of integrated, region-specific assessments that evaluate biogeophysical (albedo) and biogeochemical (carbon) effects as a single coupled system.

Progress is constrained, however, by persistent uncertainties in how these processes unfold across space and time. The magnitude and durability of albedo changes vary widely across land-use types and management practices, while the sensitivity of low-cloud regimes to surface forcing depends on the prevailing circulation state. Earth system models still tend to represent cloud-surface coupling in a fragmented manner, and long-term, high-resolution albedo records remain insufficient to resolve low-frequency variability or emerging regional trends. These limitations complicate efforts to quantify the climatic consequences of land-surface interventions and to anticipate their interactions with atmospheric reorganization.

Translating this understanding into practice requires three mutually reinforcing operational pillars: monitoring, constraint, and optimization. Monitoring depends on operational satellite products (e.g. MODIS, CERES, VIIRS successors) capable of continuously tracking cloud fields and surface reflectivity at policy-relevant scales [3], providing the radiative context needed to detect emerging anomalies. Constraint relies on proxy-based climate reconstructions that preserve low-frequency natural variability [26], anchoring modern observations within a multi-centennial baseline to distinguish structural shifts from oscillatory background behavior. Optimization involves decision-support tools that jointly evaluate carbon sequestration and surface-reflectivity outcomes [27], enabling practitioners to maximize climate benefits while minimizing unintended trade-offs across heterogeneous landscapes. Together, these elements define a research agenda that improves atmospheric representation while actively managing surface properties, a dual strategy that offers a pragmatic pathway to slow the accumulation of radiative energy.

7. Conclusion and outlook

The contraction of cloud regimes, the long-term decline of Mediterranean cloudiness, and widespread shifts in surface albedo are not isolated phenomena.

Rather, they are interconnected components of a broader transition in the Earth's energy balance: a rapidly dimming planetary mirror. As skies clear, a larger fraction of incoming shortwave radiation reaches the surface, increasing the climatic influence of land and ocean reflectivity. Rather than representing competing explanations, cloud contraction and surface albedo act as dynamically coupled controls, with changes in cloud cover determining the extent to which surface reflectivity influences the planetary radiation budget. This systems perspective is further reinforced by recent satellite observations showing that Earth's planetary albedo is maintained through compensating interactions among atmospheric circulation, cloud radiative effects, and clear-sky surface reflectivity, highlighting the inherently coupled nature of the shortwave radiation budget across hemispheric scales [28]. Under these clearer-sky conditions, surface properties exert growing leverage on regional and potentially global energy budgets. Integrating this coupling into climate science and policy offers three distinct advantages: it sharpens attribution by forcing joint consideration of circulation changes, internal variability, and surface processes; it exposes critical gaps in Earth system models, demanding better representation of both cloud dynamics and low-frequency regional shifts; and it elevates surface-albedo management from a niche concept to a vital, complementary mitigation lever alongside carbon reduction. These insights demand a comprehensive approach to climate governance that actively manages both atmospheric composition and surface reflectivity. By refining our understanding of atmospheric processes while strategically optimizing land-surface properties, society can develop more robust pathways to slow the accumulation of radiative energy. Navigating this emerging era of clear-sky amplification requires us to treat the Earth's surface not merely as a recipient of climate change, but as an active participant in its stabilization.

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Data availability statement

The data that support the findings of this study are available upon reasonable request from the authors.

The Perspective is not based original data but mainly on results from published literature.

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